

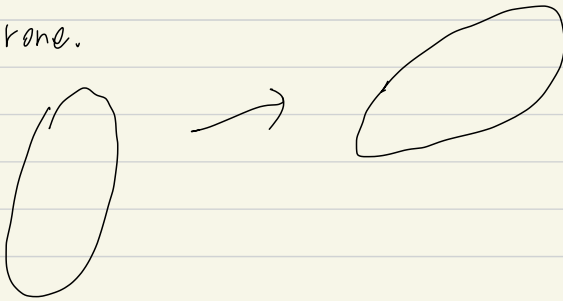
Chapter 3. Linearization

Motivation

• 3D Rotation. $SO(3)$,

• Rigid body motion $SE(3)$. (rotation + translation)

Drone.



$$SE(3) = \left\{ \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix} \mid R \in SO(3), p \in \mathbb{R}^3 \right\}$$

$$G \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} Rx + p \\ 1 \end{pmatrix}$$

$$G_1 = \begin{pmatrix} R_1 & p_1 \\ 0 & 1 \end{pmatrix} \quad G_2 = \begin{pmatrix} R_2 & p_2 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow G_1 G_2 = \begin{pmatrix} R_1 R_2 & R_1 p_2 + p_1 \\ 0 & 1 \end{pmatrix}$$

Lie group : Algebra + Calculus,
+ Geometry

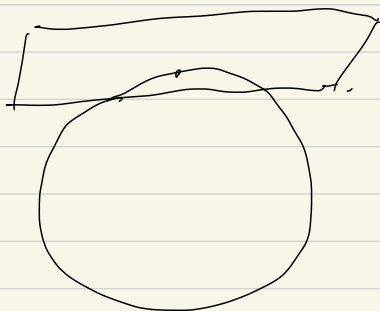
Three concept

1. Tangent plane.

2. Group homomorphism

3. Differential

3.1 Tangent plane,



classic.:

normal vector field. $\vec{n}$

$$\vec{n} \cdot (x - p) = 0$$

Matrix group $G \subseteq GL(n, \mathbb{R})$ ($\det M \neq 0$).

$$\subseteq \\ I_n.$$

Let $r(t)$: $n \times n$ matrix.

Def, "Tangent space at the identity of G ,"

$$L(G) = \{ r'(0) \mid r: \mathbb{R} \rightarrow G, r(0) = I_n, r: \text{smooth} \}$$

Ex. (3.1.6, 3.1.8) $L(G)$: vector space.

① $L(G)$: closed under scalar multiplication

② $L(G)$: closed under addition

Pf ① Let $A \in L(G)$. $A = r'(0)$ s.t. $r(0) = I$, $r(t) \in G$,
 Define $\beta(t) = r(t) \in G$, $\beta'(t) = r'(t)$ $\beta'(0) = r'(0) \in L(G)$,
 $\Rightarrow cA \in L(G)$

② Let $A, B \in L(G)$. $A = \alpha'(0)$, $B = \beta'(0)$

$$\text{Define. } r(t) = \alpha(t) \beta(t) \quad r'(0) = \alpha'(0) \beta(0) + \alpha(0) \beta'(0)$$

$$= A \cdot I + I \cdot B$$

$$= A + B \quad \Rightarrow A + B \in L(G)$$

Ex. $\{\det A \neq 0\}$

Let $G = GL(2, \mathbb{R})$ Find $L(G)$

Def) $r_1(t) = \begin{bmatrix} e^t & 0 \\ 0 & 1 \end{bmatrix}$ $r_2(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$

$r_3(t) = \begin{bmatrix} 1 & 0 \\ 0 & e^t \end{bmatrix}$ $r_4(t) = \begin{bmatrix} 1 & 0 \\ t & 1 \end{bmatrix}$

$r_i(0) = I$

$r_1'(0) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ $r_2'(0) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$r_3'(0) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ $r_4'(0) = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

$r_i'(0) \in L(G)$ ($L(G)$ closed under scalar mult & addition)

$\Rightarrow L(G) = M(2, \mathbb{R})$

$$(\det A = 1)$$

Ex 2) Let $G = SL(2, \mathbb{R})$ Find $L(G)$

pf)

$$\text{Let } v(t) = \begin{bmatrix} a(t) & b(t) \\ c(t) & d(t) \end{bmatrix}$$

$$v(0) = I_2 \Rightarrow a(0) = 1 \quad b(0) = 0 \quad c(0) = 0 \quad d(0) = 1$$

$$v(t) \in G \Rightarrow ad - bc = 1.$$

$$a'd + ad' - b'c - bc' = 0.$$

$$\text{at } t=0 \Rightarrow a'(0) + d'(0) = 0.$$

Claim

$$L(G) = \left\{ \begin{bmatrix} a & b \\ c & -a \end{bmatrix} ; a, b, c \in \mathbb{R} \right\}$$

$$v_1(t) = \begin{bmatrix} e^t & 0 \\ 0 & e^{-t} \end{bmatrix} \quad v_2(t) = \begin{bmatrix} e^t & t \\ 0 & e^{-t} \end{bmatrix} \quad v_3(t) = \begin{bmatrix} e^t & 0 \\ t & e^{-t} \end{bmatrix}$$

$$v_1'(0) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad v_2'(0) = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \quad v_3'(0) = \begin{bmatrix} 1 & 0 \\ 1 & -1 \end{bmatrix}$$

3.2 Group homomorphism

Symmetry

Def. A symmetry of a group G_1 with binary operation $*$ is an invertible function $F: G_1 \rightarrow G_1$.

$$\text{s.t. } F(q * h) = F(q) * F(h) \quad \forall q, h \in G_1$$

Def $F: G_1 \rightarrow G_2$ is Group homomorphism if.

$$F(q * h) = F(q) * F(h) \quad \forall q, h \in G_1 \quad (3.1)$$

Def $F: G_1 \rightarrow G_2$ is Group Isomorphism if

F is one to one & onto satisfying (3.1)

$$\det M = 1.$$

Ex.) Define $F: SL(n, \mathbb{R}) \rightarrow SL(n, \mathbb{R})$ by $F(X) = (X^{-1})^T$.

Prove F is isomorphism

$$\det X = 1 \quad \det X^{-1} = 1, \quad \det (X^{-1})^T = 1.$$

$$\Rightarrow F(X) \in SL(n, \mathbb{R})$$

$$\begin{aligned} F(XY) &= ((XY)^{-1})^T = (Y^{-1}X^{-1})^T = (X^{-1})^T (Y^{-1})^T \\ &= F(X) F(Y) \end{aligned}$$

(cont.) Let $\gamma \in SL(n, \mathbb{R})$ $(\gamma^T)^{-1} \in SL(n, \mathbb{R})$

(f.i) If $\begin{matrix} \gamma_1 & = & \gamma_2 \\ \parallel & & \parallel \\ F(\gamma_1) & & F(\gamma_2) \end{matrix} \quad \Rightarrow \quad (\gamma_1^{-1})^T = (\gamma_2^{-1})^T \Rightarrow \gamma_1 = \gamma_2$

$$F^{-1}(\gamma) = (\gamma^T)^{-1}$$

Ex 2), $M^T C M = C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
 $M \in O(n, \mathbb{R})$ & $\det M = 1$

Let $G_1 = SO(2, \mathbb{R})$, $G_2 = S^1 = \{ e^{i\alpha}; \alpha \in \mathbb{R} \}$

define $F: G_1 \rightarrow G_2$

$F\left(\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}\right) = e^{i\alpha}$. Prove that
 F is isomorphism

$$F\left(\begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix}\right)$$

$$= F\left(\begin{bmatrix} \cos \alpha + \beta & -\sin \alpha + \beta \\ \sin \alpha + \beta & \cos \alpha + \beta \end{bmatrix}\right) = e^{i(\alpha + \beta)} = e^{i\alpha} e^{i\beta}$$

3.3 Differentials

Homomorphism $F: G_1 \rightarrow G_2$

$$F: S_1 \longrightarrow S_2$$

$$r_1(t_0) = p_1 \longrightarrow r_2(t_0) = p_2 \quad r_2(t) = F(r_1(t))$$

" $F(p_1)$

Def differential of F at p_1

$$dF[r_1'(t_0)] = r_2'(t_0) \quad \left(\begin{array}{l} r_1(t_0) = p \\ F(r_1(t)) = r_2(t) \end{array} \right)$$

Ex. $F(x, y) = (f_1(x, y), f_2(x, y))$

$$dF[r_1'(0)] = r_2'(0) \quad \text{where} \quad r_2(t) = F(r_1(t))$$

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}$$

Ex) Let $S_1 = \{(x, y, z) : z = \sqrt{x^2 + y^2}\}$

Choose $p_1 = (3, 0, 3)$, $F(x, y, z) = (y, x, z^2)$

Let $r_1(t) = (t, 0, t)$ compare $dF|_{r_1(t)} = r_2'(t)$

$\Rightarrow S_2 = F(S_1) = (y, x, x^2 + y^2)$

$r_2(t) = F(r_1(t)) = (0, t, t^2)$

$r_1'(t_0) = (1, 0, 1)$

$r_2'(t_0) = (0, 1, 6)$

$dF = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 2x & 2y & 0 \end{bmatrix}$

$dF|_{p_1} r_1'(t_0) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 6 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 6 \end{bmatrix} = r_2'(t_0)$

$$(\det M = 1)$$

$$\text{Ex)} \quad \text{Let } F: SL(2, \mathbb{R}) \rightarrow SL(2, \mathbb{R}) \quad F(X) = (X^{-1})^T.$$

(F : group isomorphism)

$$X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \Rightarrow X^{-1} = \begin{bmatrix} x_{22} & -x_{12} \\ -x_{21} & x_{11} \end{bmatrix}$$

$$(X^{-1})^T = \begin{bmatrix} x_{22} & -x_{21} \\ -x_{12} & x_{11} \end{bmatrix}$$

$$F(x_{11}, x_{12}, x_{21}, x_{22}) = (x_{22}, -x_{21}, -x_{12}, x_{11})$$

$$J = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Let } A \in LG, \quad \left(\text{recall } LG = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \right)$$

$$A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} = (a \ b \ c \ -a)$$

$$dF(A) = J A = (-a \ -c \ -b \ a)$$

$$= \begin{bmatrix} -a & -c \\ -b & a \end{bmatrix},$$

w.r.t $F(x) = (x^T)^T$

$$= (-A)^T, \quad \text{--- tangent space}$$

Ex.) Let $G_1 = GL(2, \mathbb{R})$, $G_2 = (\mathbb{R} \setminus \{0\}, \cdot)$.

$$F: GL(2, \mathbb{R}) \rightarrow G_2, \quad F(X) = \det X,$$

$$\text{show that } dF|_{I_2}: M(2, \mathbb{R}) \rightarrow \mathbb{R},$$
$$\Downarrow$$
$$X \mapsto \text{tr}(X)$$

Pf) $F: \mathbb{R}^4 \rightarrow \mathbb{R}$

$$\text{Let } X = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = (x_{11}, x_{12}, x_{21}, x_{22})$$

$$\det X = x_{11}x_{22} - x_{12}x_{21}$$

$$dF|_X = (x_{22}, -x_{21}, -x_{12}, x_{11}),$$

$$dF|_I = (1, 0, 0, 1),$$

$$dF|_I \cdot X = x_{11} + x_{22} = \text{tr}(X),$$

$$F: GL(n, \mathbb{R}) \rightarrow GL(1, \mathbb{R}), \quad F(X) = \det X,$$

$\Rightarrow F$: homomorphism

$$dF|_{I_n}: M(n, \mathbb{R}) \rightarrow M(1, \mathbb{R}), \quad dF|_{I_n} = \text{tr}.$$

Ex) $dF|_{I_n}: L(G_1) \rightarrow L(G_2) : \text{linear transform}$

$$A_1 = \alpha_1'(0) \quad F(\alpha_1(t)) = \alpha_2(t) \quad dF(\alpha_1'(0)) = \alpha_2'(0)$$

$$B_1 = \beta_1'(0) \quad F(\beta_1(t)) = \beta_2(t) \quad dF(\beta_1'(0)) = \beta_2'(0)$$

$$\text{Let } r_1(t) = \alpha_1(t) \beta_1(t)$$

$$F(r_1(t)) = r_2(t)$$

$$dF(\alpha_1'(t) \beta_1(t) + \alpha_1(t) \beta_1'(t)) = r_2'(t)$$

$$dF|_{I_n} (\alpha_1'(0) I + I \beta_1'(0)) = r_2'(0)$$

$$= \alpha_2'(0) + \beta_2'(0)$$

$$\Rightarrow dF(A_1 + B_1) = dF(A_1) + dF(B_1)$$

$$(dF(cA) = c dF(A))$$



$$\text{Let } G_1 \subseteq GL(n, \mathbb{R}), \quad G_2 \subseteq GL(m, \mathbb{R})$$

$$\& \quad F: G_1 \rightarrow G_2 \quad \text{homomorphism}$$

Then F is called Lie group homomorphism

$$F: G_1 \rightarrow G_2$$

$$\begin{aligned} dF: & \mathcal{L}(G_1) \\ & \hookrightarrow \\ & A_1 \end{aligned}$$

$$r_1: \mathbb{R} \rightarrow G_1 \quad r_2 = F(r_1(t))$$

$$r_1(0) = I_n$$

$$r_1'(0) = A_1$$

$$dF(A_1) = r_2'(0)$$